

Fishing Ground Identification from Mexican VMS Data

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1 Methods

We identified fisher-level fishing grounds from Mexican Vessel Monitoring System (VMS) data using DBSCAN density-based clustering. The analysis covers 5 gear types: shrimp trawl, set longline, drifting longline, small pelagic purse seine, and tuna purse seine. Raw VMS pings were filtered by gear-specific speed thresholds and a minimum distance of 1 km from shore.

minPts (domain-based rule). We define a fishing ground as a location with at least 5 hours of fishing activity on at least 5 distinct days per year. Since VMS pings are recorded at approximately hourly intervals, this translates to a base threshold of 25 pings per vessel-year of data. For each vessel, $\text{minPts} = 25 \times N_{\text{years}}$, where N_{years} is the number of distinct calendar years in the vessel’s track. This scaling ensures that vessels with longer records must demonstrate sustained, repeated use of a location for it to qualify as a fishing ground.

eps (kNN elbow method). The neighbourhood radius was calibrated per gear type using per-vessel k -nearest-neighbour distance curves with $k = 25$. For each qualifying vessel (those with ≥ 50 pings), we computed the sorted k -NN distance curve and detected the elbow via the simplified kneedle algorithm (maximum perpendicular distance from the line connecting the first and last points). The gear-level eps is the median of all per-vessel elbow estimates.

Table 1: DBSCAN parameters by gear type. eps is the gear-level median of per-vessel kNN elbow estimates. minPts varies per vessel (shown as median across vessels in each gear).

Gear	eps (km)	Median minPts	Range minPts
Drifting longline	17.7	300	25–475
Set longline	12.4	350	25–425
Shrimp trawl	9.1	400	25–475
Purse seine	12.3	375	100–475
Tuna purse seine	141.0	400	25–475

After clustering, each cluster’s convex hull defines a fishing ground. We computed ground area, total fishing hours, number of pings, and number of distinct visit dates per ground.

2 Results

We identified **4,651** fishing grounds across **1172** vessels (Table 2). Figure 1 shows the spatial distribution of identified grounds.

Table 2: Summary of identified fishing grounds by gear type.

Gear	Vessels	Total grounds	Median grounds/vessel	Median area (km ²)	Median hours
Drifting longline	40	148	4	22764.6	6746.7
Set longline	143	364	2	4980.3	3234.0
Shrimp trawl	862	3800	4	6402.3	11937.4
Purse seine	78	257	3	3524.4	4027.5
Tuna purse seine	49	82	1	164997.7	1413.3

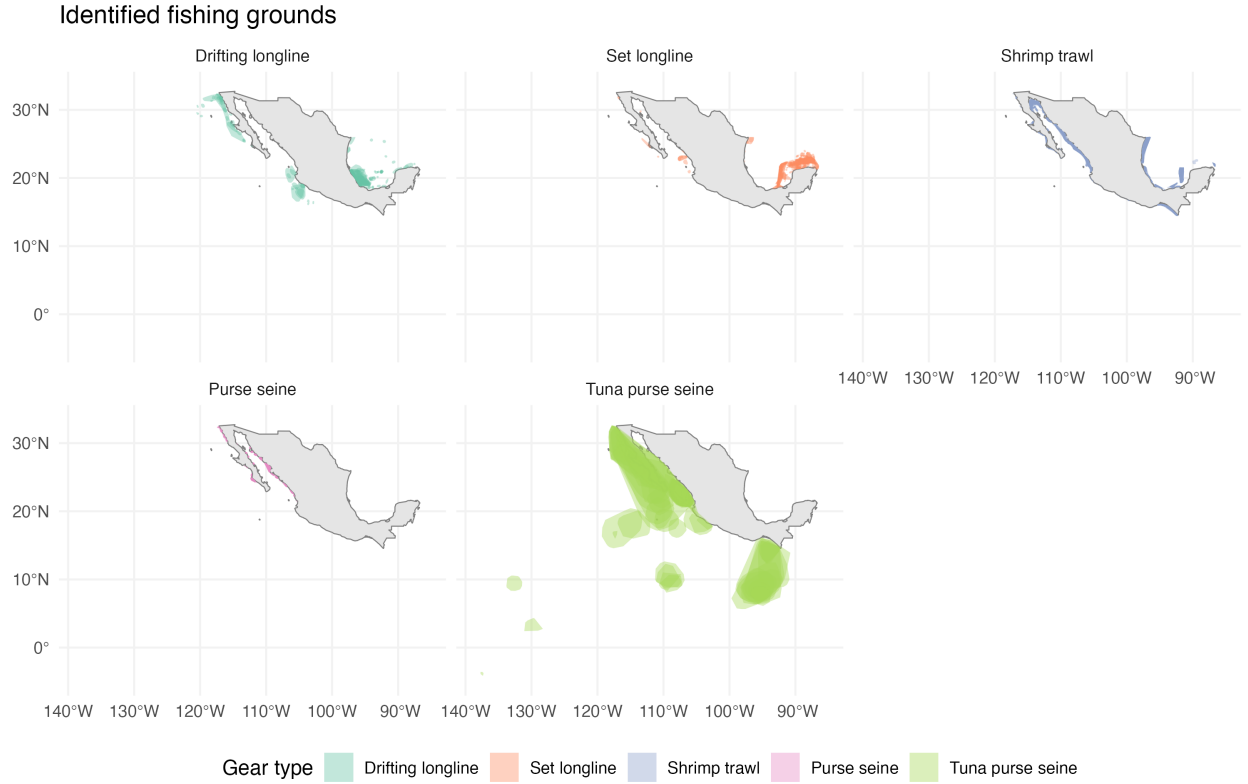


Figure 1: Spatial distribution of identified fishing grounds by gear type.

Figure 2 shows the distribution of total fishing area and number of grounds per vessel. Most vessels operate across a modest number of grounds, though the range varies considerably by gear.

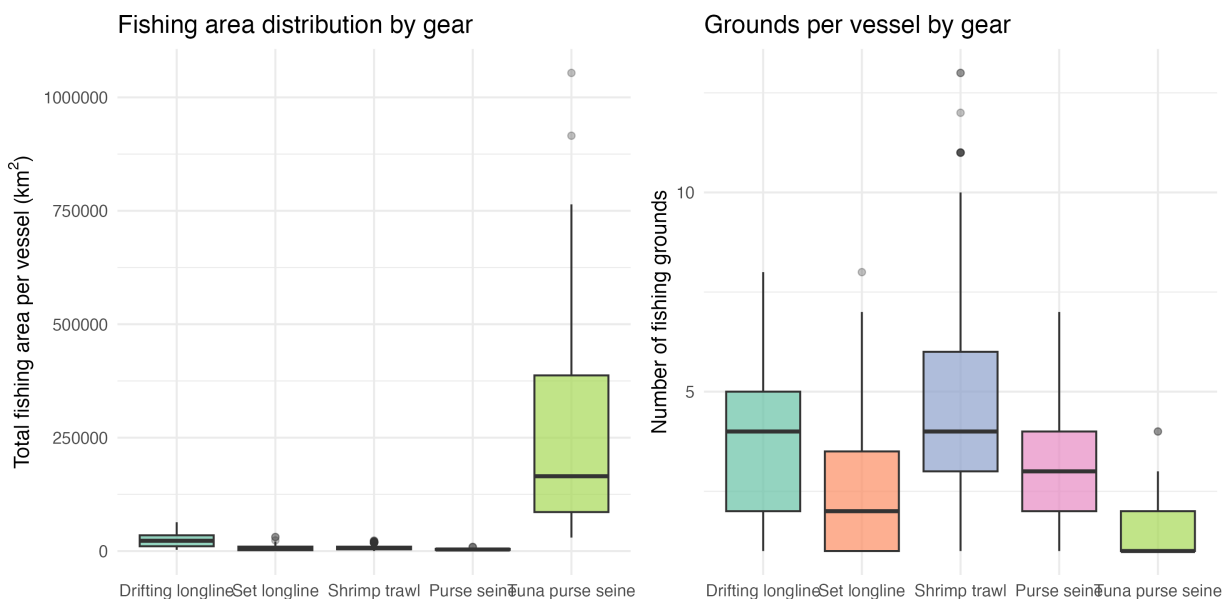


Figure 2: Total fishing area (left) and number of grounds (right) per vessel, by gear type.

Figure 3 shows how the domain-based minPts rule distributes across vessels in each gear type. The spread reflects differences in how many years of VMS data each vessel has.

Figure 4 shows the per-vessel eps estimates used to calibrate the gear-level neighbourhood radius.

Figure 5 characterises how frequently vessels return to their identified grounds and the typical time between revisits.

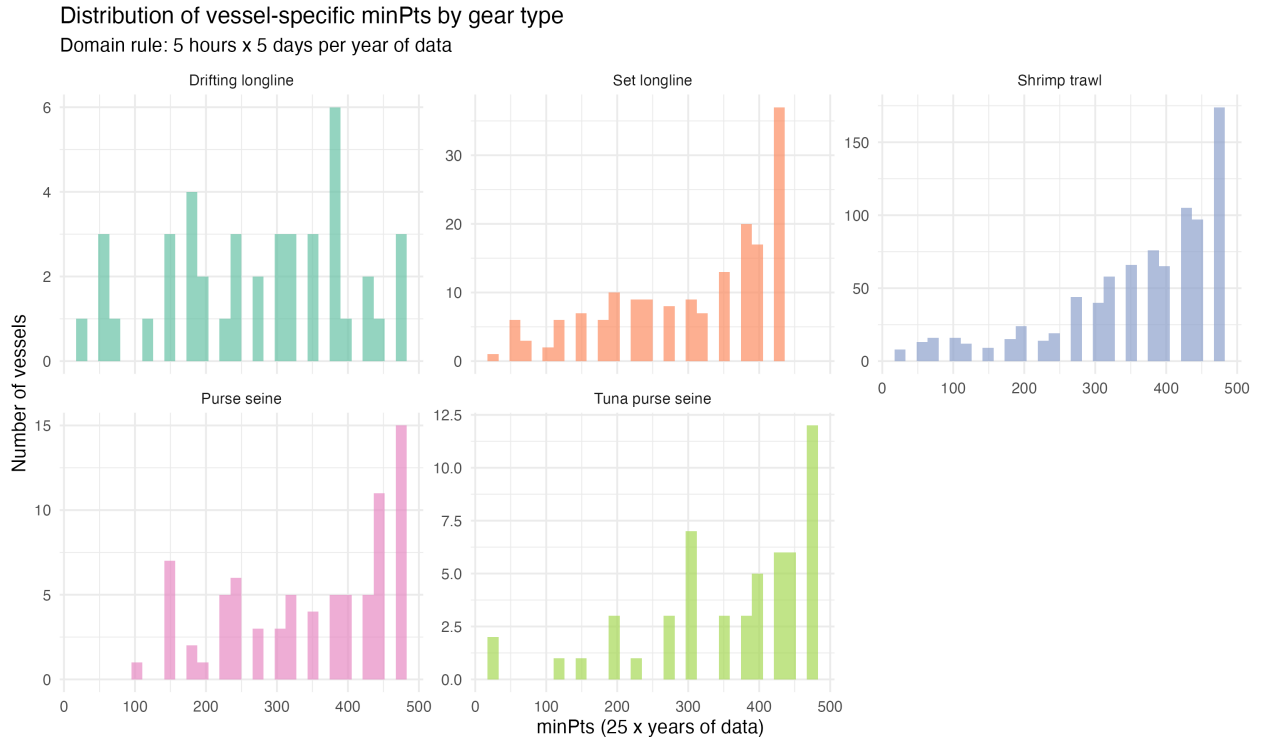


Figure 3: Distribution of vessel-specific minPts values. Longer-tracked vessels require more pings to form a ground.

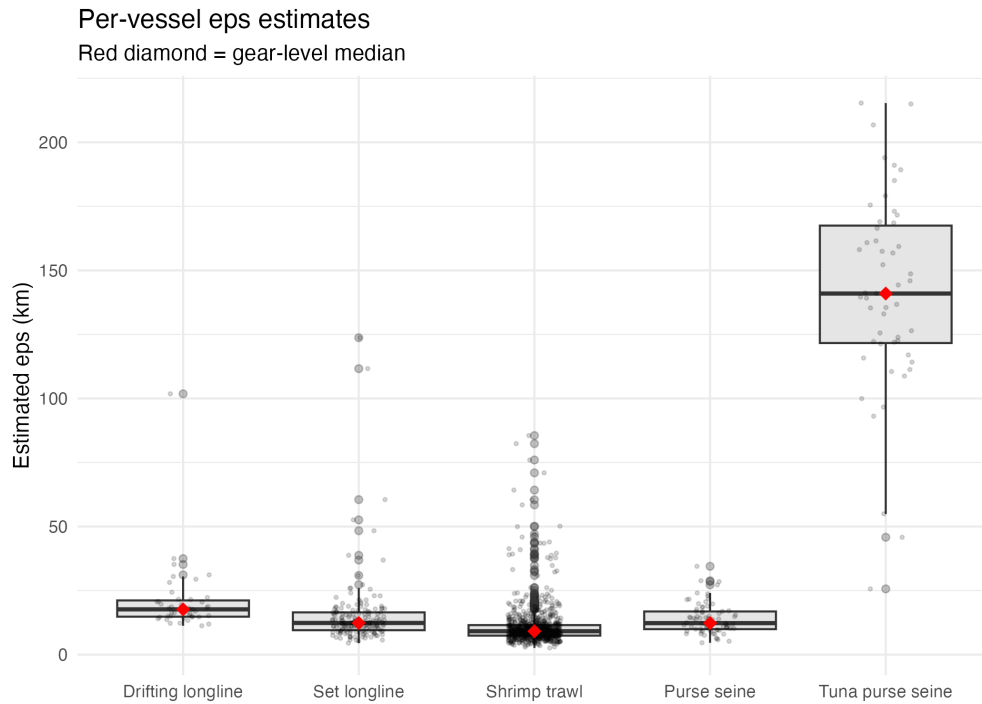


Figure 4: Per-vessel eps estimates by gear. The red diamond marks the gear-level median used for clustering.

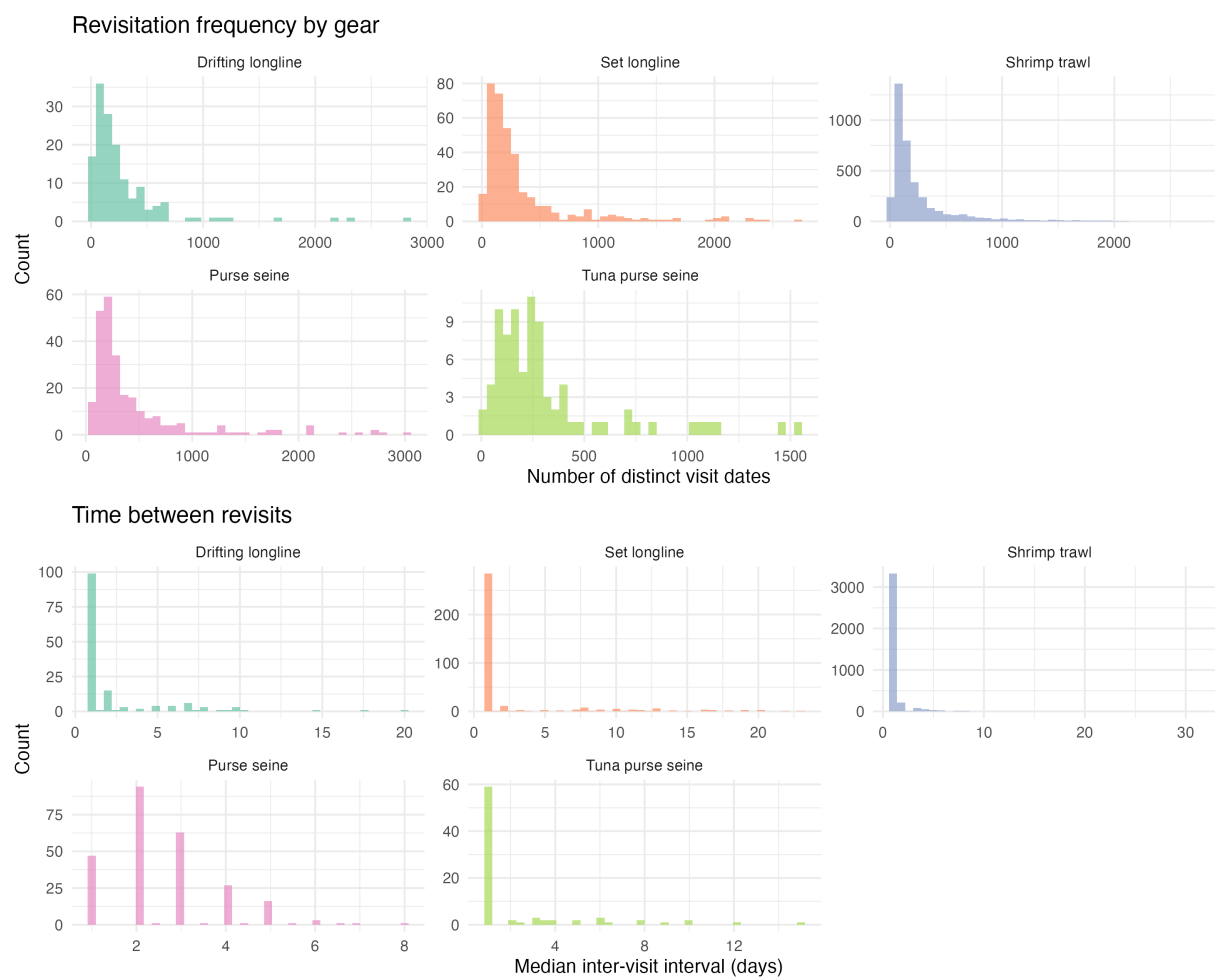


Figure 5: Revisitation patterns: number of distinct visit dates per ground (top) and median days between visits (bottom).

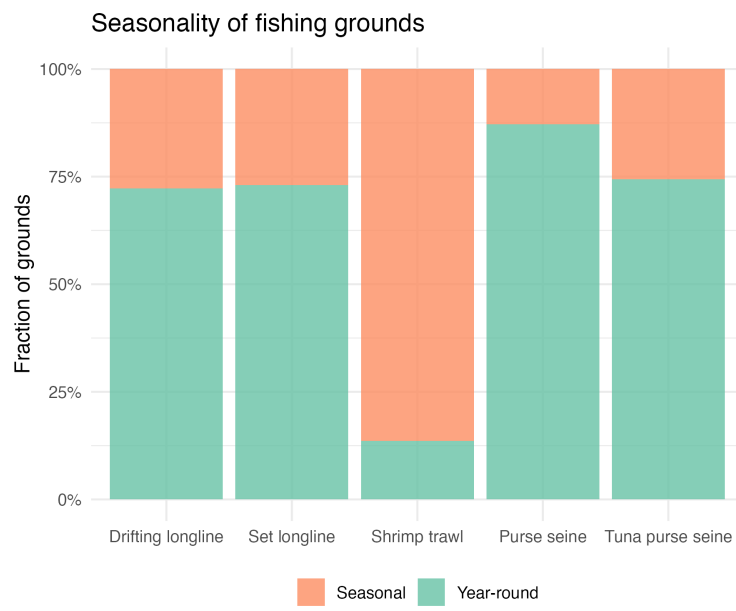


Figure 6: Fraction of fishing grounds classified as seasonal vs. year-round by gear type.